Kshitij Satish Mahajan

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Summary

MBA candidate with an undergraduate foundation in Data Science, bridging technical rigor and business decision-making. Author of three first-author IEEE-published papers spanning statistical hypothesis testing, ML regression and a critical evaluation exposing a false-precision flaw in AQI forecasting models. Proficient in Excel, Python, SQL and Power BI for end-to-end analysis.

Technical Skills

Languages: Python (Data Analysis, Machine Learning, Visualization), SQL
Tools & Platforms: Power BI, Microsoft Excel, Google Colab, Jupyter Notebook, Visual Studio Code, Git
Areas of Proficiency: Data Science, Machine Learning, Model Building & Evaluation, Data Visualisation, Dashboard Design, Insight presentation and Report Creation

Education

Goa Institute of Management — Panjim, Goa
Post Graduate Diploma in Management (Big Data Analytics) — *2026 - 2028*

RV University — Bengaluru, Karnataka
Bachelor of Science in Data Science; CGPA: 8.84 — *2023 – 2026*

Publications & Projects

Limitations of Predictive Modeling for AQI Forecasting — January 2026

- Proved ML-based AQI forecasting is fundamentally misapplied by showing models merely reverse-engineer CPCB's deterministic sub-index formula rather than learning genuine patterns: same-day accuracy of R^2=99.28% (XGBoost) collapsed to R^2=50.34% under a 7-day lag across 366 days of Bengaluru air quality data. Presented at IITCEE; published in IEEE Xplore.

Child Mortality and Healthcare Expenditure Analysis using ML & Statistical Methods — January 2026

- Quantified an 89% gap in child mortality (0.58 vs. 5.39 deaths/100 live births) between high- and low-healthcare-expenditure nations across 189 countries using Welch's t-test ($t=14.38$, $p<0.001$) and ANOVA ($F=138.07$, $p=1.74\times10^{-37}$), validated through K-means/GMM clustering. Demonstrated how integration of statistical inference and unsupervised ML supports data-driven healthcare policy and resource allocation. Presented at ICEI; published in IEEE Xplore.

Sustainable Water Management using Machine Learning Regression — November 2024

- Built XGBoost and Random Forest models on Telangana government infrastructure data (189 cities, 2022) to predict urban water demand, achieving MAE=0.91 MLD (R^2=91.78%) with bootstrapped validation, and residential connection counts accurate to within 0.0003–0.0041 (10th–90th percentile error). Presented at CSITSS; published in IEEE Xplore.

Experience

Data Science Intern - Dhee Centre for AI and Data Science — Bengaluru - June 2024

- Applied XGBoost and Random Forest regression to a Telangana government water infrastructure dataset; work was peer-reviewed and published as a first-author paper in IEEE Xplore (see Publications)

Certifications & Achievements

IBM Data Science Professional Certificate *(View Credential)*: Gained hands-on experience in Python, data analysis and visualization. Involved a capstone project.
1st Place - Analytica 3.0, Intercollege Data Analysis Competition

Community Service

Student Volunteer - Youth For Parivarthan NGO — Bengaluru - 2025

- Restored public spaces through cleaning/repainting drives and led a book collection initiative for recycling and redistribution to government schools.